Building a Standardized Database for Honey Bee Microbiome: Addressing Metadata and Data Comparability gaps.

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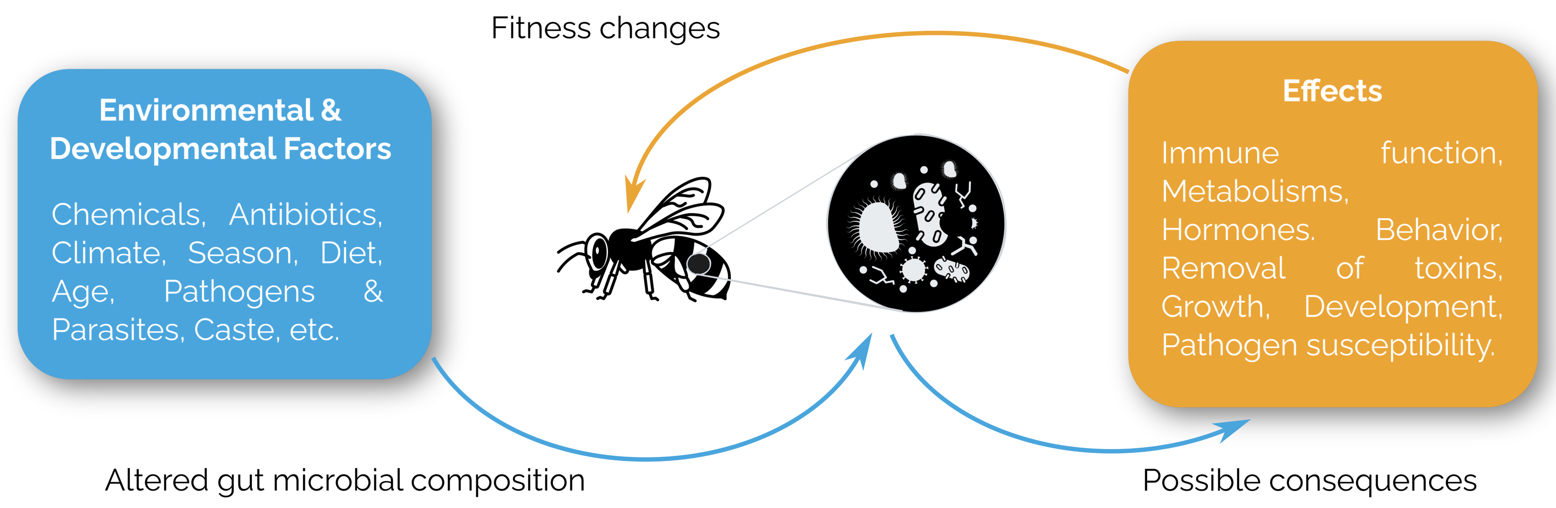
Background

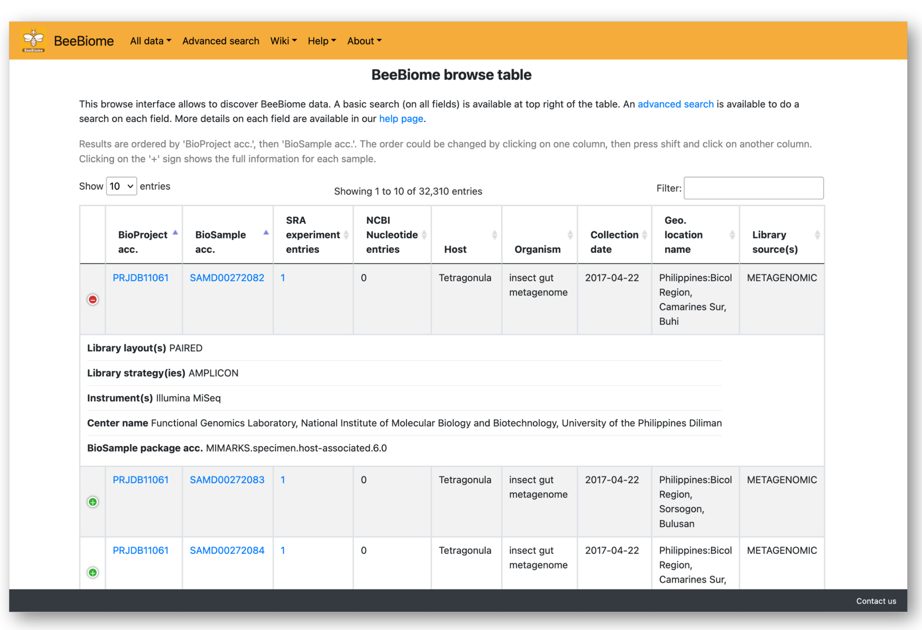
Environmental and developmental factors influence the gut microbiota of bees, leading to fitness changes and various effects such as altered immune function, metabolism, and development.



Adapted from Rayman & Moran, Curr Opin Insect Sci, 2018 (10.1016/j.cois.2018.02.012)

Problem

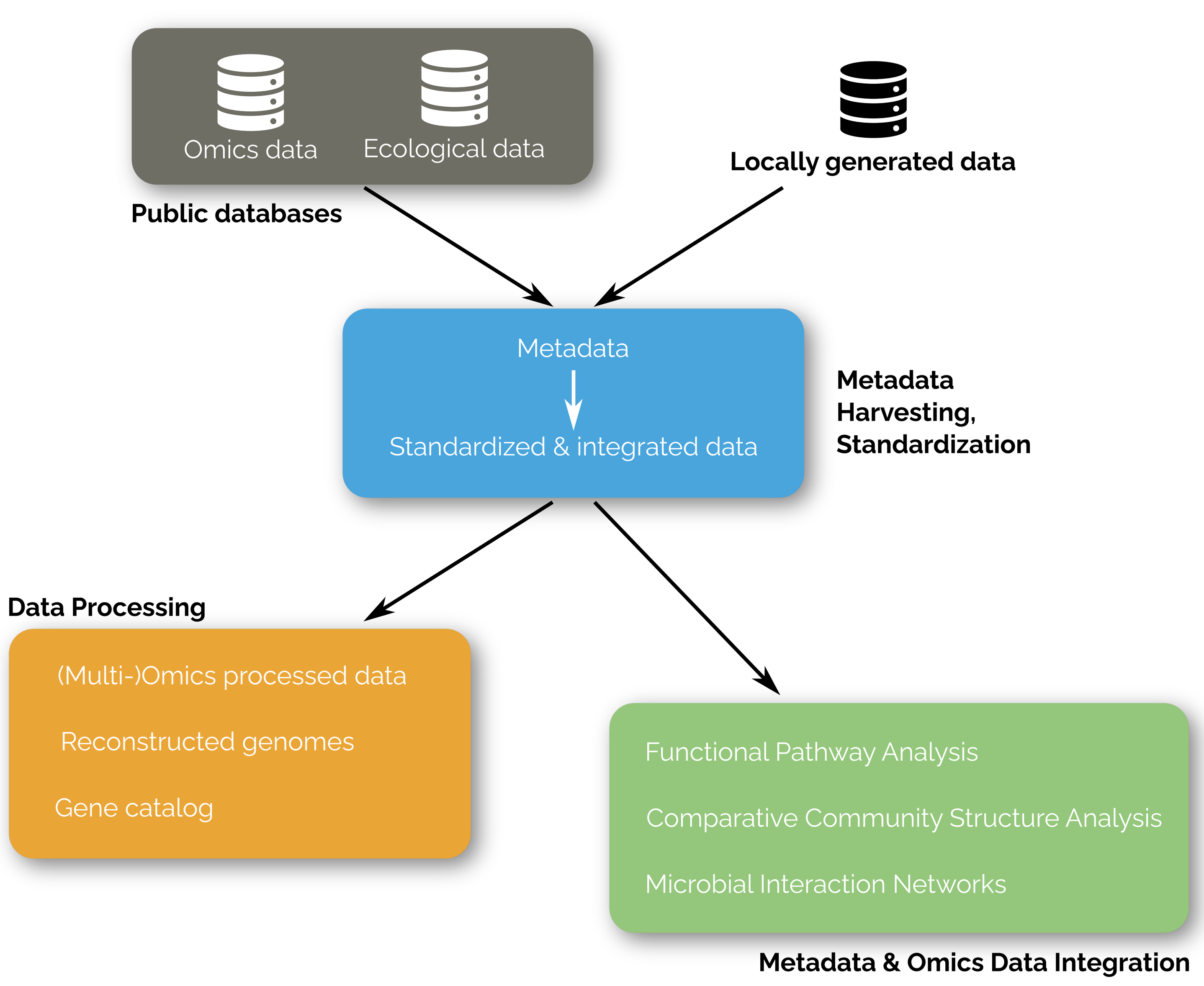
Recent advances in metagenomic sequencing have generated extensive datasets for studying the honey bee microbiome. However, challenges remain in terms of (meta)data accessibility and comparability due to varying analysis methods and limited integration across omics layers.



BeeBiome (beebiome.org), an international consortium, offers a catalog of over 34,000 datasets on bee microbiomes. This catalog faces issues with sparse metadata and lack of comparable, analyzed data, limiting holistic and reproducible analyses.

Solution

To tackle these challenges, we have launched a collaborative effort between the Laboratoire Microorganismes: Génome et Environnement (LMGE) and Auvergne Bioinformatique (AuBi) Platform.
Our goal is to develop BeeBiome 2.0, a cutting-edge bee microbiome database that seamlessly integrates metagenomic data with standardized and curated metadata.

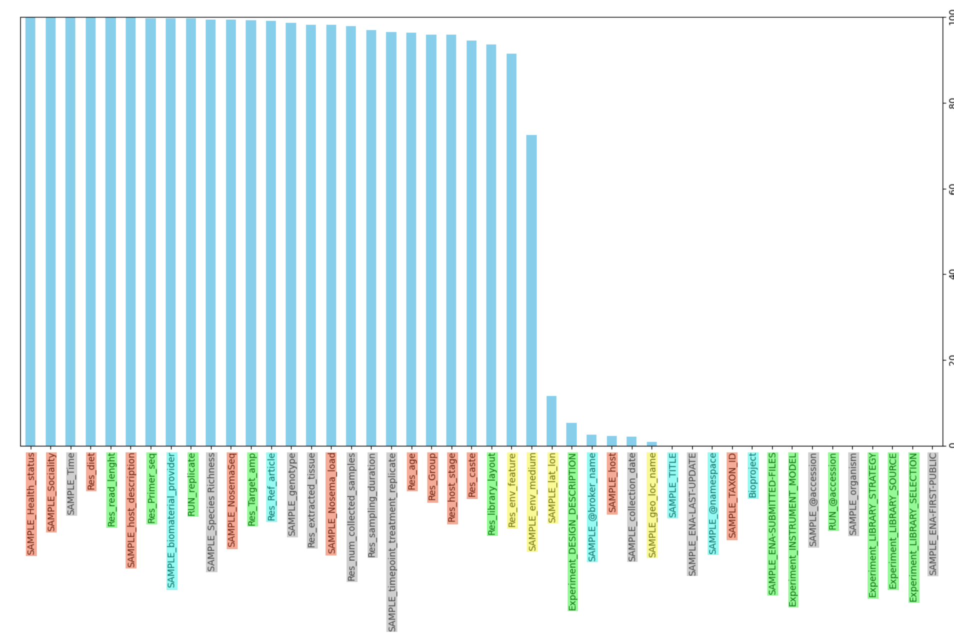


By harmonizing data formats and analysis pipelines using Galaxy, we aim to create a robust resource for the scientific community, enabling more accurate and reproducible research on honey bee health.

Preliminary Results & Challenges

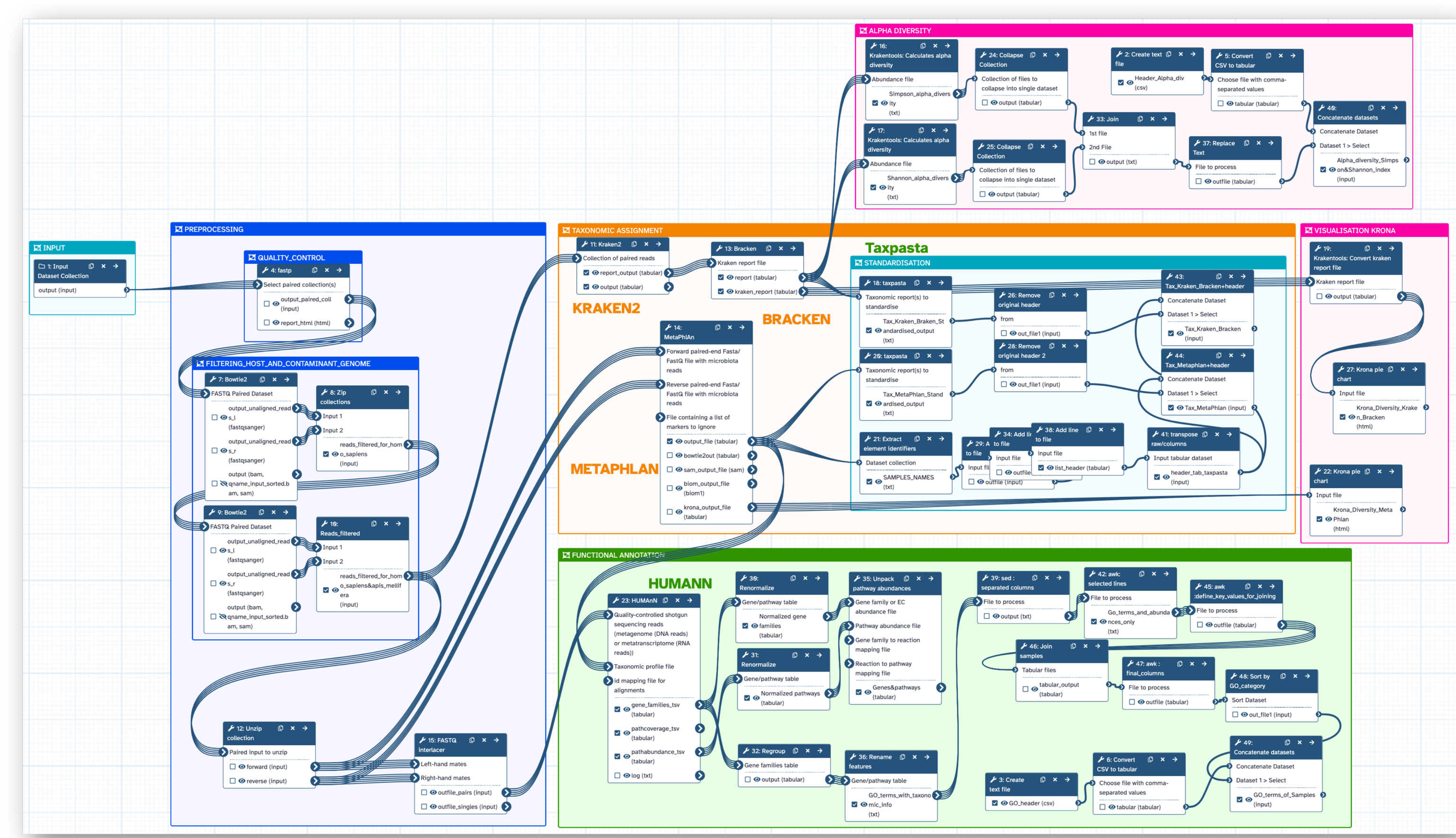
Preliminary works including two master's internships supervised in 2024 by both LMGE researchers and AuBi research engineers have already laid the groundwork, focusing on aggregating (meta)data and developing a standardized computational Galaxy pipeline.

Metadata heterogeneity: Many of missing values for the metadata of the collected 11,788 samples from NCBI.



Poster + Links

Galaxy Workflow for Metagenomic Taxonomy and Functional Analysis, extending previous workflows



Our results highlight two key challenges hindering cross-study comparability:
(1) the lack of detailed metadata (e.g., pesticide exposure) and
(2) the needs of standardized workflows for reanalyzing datasets.

Future Directions

Metadata:
- Collaborations with microbiology and biodiversity metadata groups
- Development of methods to fill in missing or curate metadata

Data analysis
- Collaborations with the microGalaxy community to develop reproducible workflows
- Implement multi-omics and cross-study comparisons.

Database: Make it openly accessible.

Conclusions

This project aims to build a robust bee microbiome database to improve cross-study comparisons and enhancing our understanding of environmental impacts on honey bee health. The database uses advanced data analysis hosted at the Université Clermont Auvergne Mesocenter DC infrastructure. This project is a good example of a **successful collaboration** between a research lab (LMGE) and a bioinformatic plateform (AuBi)

Acknowledgments

Université Clermont Auvergne · AuBi Auvergne Bioinformatique · IFB Institut Français de Bioinformatique · LMGE

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